

Prediction Market Mechanisms

Theory and C++

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1 Introduction

Eventually, I want these notes to be the distilled map of what I have learned in researching prediction markets. The notes are ordered as I explored these topics. Here are the two objectives of the document:

- **Goal 1:** Build a compressed, intuitive representation of the mathematics behind prediction markets.
- **Goal 2:** Use C++ to learn via implementation.

2 Preliminary Intuitions

If you're reading this, you probably know how to *use* a prediction market. Maybe you've used Kalshi, where you can use USD to buy a share of a **future**. But in case you haven't used one before, you might wonder what it means to buy a share of a future. The idea is simple: you agree to pay the **market maker** (like Kalshi) 70 cents if you think there . And if it does happen, you get

3 Mathematical Setup: Scoring Rules and Forecasters

A prediction market is a mechanism that generates a price. Specifically, it generates a probability that an event occurs. Let's take an event, Stephen Miller saying the word "iron" in his next press conference. Our mechanism generates the probability .3 of that happening—a bettor can buy one share of that **future** for 30 cents. If he says "iron," they get a dollar back; if not, they walk away 30 cents poorer. But real world events are messy and many-dimensional—how might you systematically generate a price for an event occurring?

3.1 The Efficient Market Hypothesis (EMH)

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- **Strong EMH:** The price of an item determined by all the information that the public knows about it, as well as all the information that private entities (like insiders) know about it. Even if you have inside information, you can't systematically beat the market because *the information you know is reflected in the price*.
- **Semi-strong EMH:** The price of an item is determined *only by public information*. Public information like news,

3.2 Scoring Rules

3.3 Online Machine Learning Connection: Cross Entropy and KL Divergence

3.4 A General Scoring Rule Template

3.5 Forecasters

4 Making it a Market

4.1 Market Scoring Rules

A market scoring rule is really just a way to *subsidize* a scoring rule into a market: the market maker takes on the roll of the scoring rule and the traders update the price.

4.2 LMSR Cost Function

4.3 No-Regret Learning Equivalence

4.4 Calibration

Make sure this is tightly linked to the bound given in Abernethy et al. The market prices converge to calibrated prices, **even with adversarial traders**.

5 Information Aggregation

5.1 Rational Expectations Equilibrium

5.2 Grossman-Stiglitz Paradox

5.3 Brier Decomposition

6 Implementation